A Big Data Framework for Price Tracking and Product Matching in Sri Lankan E-Commerce

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Contents

Chapter 1

Evaluation and results

This chapter of the report presents the evaluation of the performance and the cost-effectiveness of each of the modules of the proposed system. The evaluation is based on the results obtained from the experiments and tests conducted on the system, and the analysis of the data collected during the experiments. These results are compared against the initial aims, objectives, and requirements of the system, in order to determine the extent to which the system meets the desired performance and cost-effectiveness criteria. Finally, they are compared against the results of similar systems in the literature, in order to assess the relative performance and cost-effectiveness of the proposed system.

1.1 Distributed web crawler

The performance of the distributed web crawler is dependent on three main factors:

- The number of URLs each crawler instance is assigned to crawl.
- The rate at which new crawler instances are spawned.
- The error rate of the crawling process.

During the testing phase, the following settings were used to evaluate the performance of the distributed web crawler:

- Each crawler instance was assigned to crawl 100 URLs.
- The rate at which new crawler instances were spawned was set to 1 instance every 3 minutes (20 instances per hour) - across 3 AWS regions.

During a period of 24 hours, the distributed web crawler was able to successfully crawl a total of 29,809 unique URLs from a collection of 36 websites, which represents an average of 1,242 URLs crawled per hour. Table **??** shows the total number of pages crawled per domain, as well as the number of pages that were updated within the last 24 hours.

Domain	Total pages	Updated in 24h
abansit.lk	286	129
barclays.lk	2876	841
bigdeals.lk	53606	9884
brownsdeals.com	795	184
buyabans.com	10363	3509
buyzone.lk	308	75
celltronics.lk	2508	998
chamacomputers.lk	3360	1450
damro.lk	1662	549
dialcom.lk	577	174
finez.lk	409	118
fireworks.lk	1608	594
gamestreet.lk	2640	1206
geniusmobile.lk	680	270
homelux.lk	239	57
idealz.lk	236	39
laptop.lk	2000	715
lifemobile.lk	3497	0
luxuryx.lk	1266	185
mansafit.com	713	93
mcentre.lk	721	294
mysoftlogic.lk	762	174
nanotek.lk	2219	674
onei.lk	5161	1962
pettahkade.lk	545	154
raesl.lk	315	49
redlinetech.lk	3154	380
rooter.lk	293	150
simplytek.lk	3127	1383
singersl.com	4328	1357
singhagiri.lk	817	199
strong.lk	349	91
techzone.lk	1998	964
ugreen.lk	1251	330
xclusive.lk	1198	242
xmobile.lk	1095	336
TOTAL	**116,962**	**29,809**

Table 1.1: Per-domain crawl totals and URL updates within the last 24 hours.

During this same period, there were a total of 972 errors encountered by the distributed web crawler, which represents an error rate of approximately 3.26% of the total URLs crawled. However, many of these URLs were crawled successfully on retry. Of which,

- 324 were timeout errors, which represents instances where the crawler was unable to establish a connection to the target website and render the webpage within the specified timeout period.
- 458 were HTTP 403 errors, which indicates that the crawler was denied access to the target website by the website's server.
- 42 were HTTP 404 errors, which indicates that the requested URL was not found on the target website.
- 30 were HTTP 429 errors, which indicates that the crawler was rate-limited by the target website due to too many requests being sent in a short period of time.

Out of the total of 116,962 unique URLs crawled, 72,899 were classified as product pages, while 44,063 were classified as non-product pages. If only the product pages are considered, within the last 3 days, a total of 71,241 URLs were crawled, which means that 1,658 pages missed the 72-hour update requirement. Of these, 1,651 were from a single domain (`lifemobile.lk`), which seems to have implemented aggressive anti-bot measures, preventing the crawler from accessing and crawling the product pages on their website.

Figure **??** shows the velocity of the distributed web crawler over a period of 30 days, which represents the number of unique URLs in 30-minute intervals that were successfully crawled and stored in the database along with a moving average of the velocity over a 24-hour period. It also shows the effect of the anti-bot measures implemented by CloudFlare, and the steps taken to mitigate the impact of these measures on the crawling process, which includes reducing the crawling rate and implementing anti-bot measures.

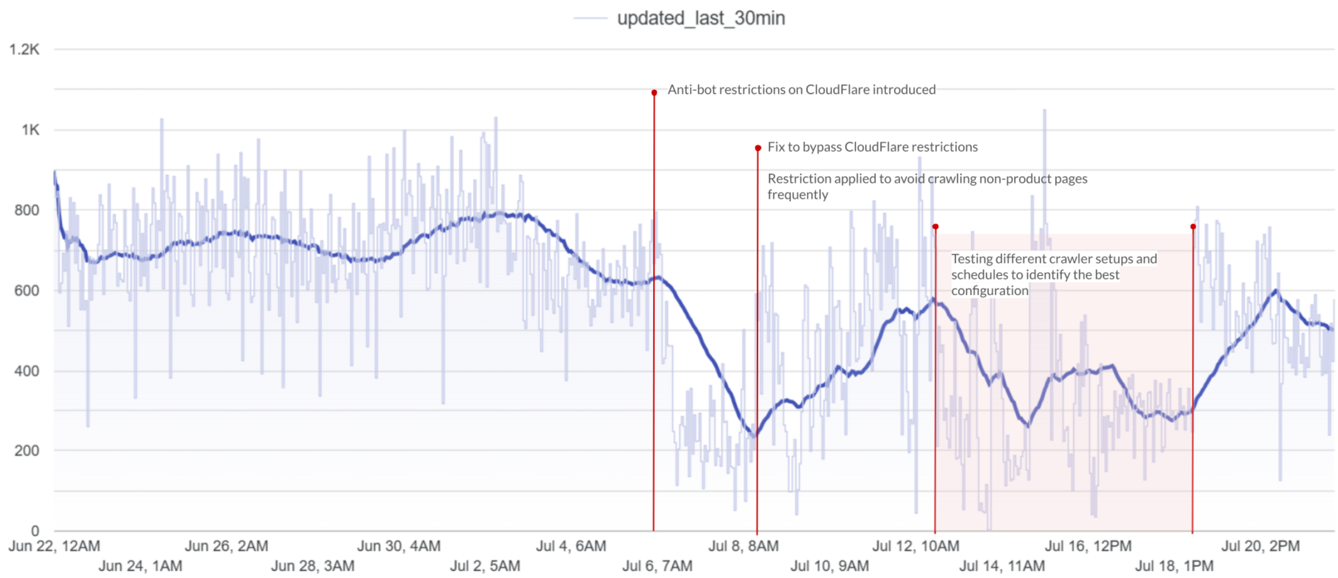

Figure 1.1: Distributed crawler velocity over 30 days

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